Chapters 1–5 Warm-Up: Python Review

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1 Expressions and Variables

1. Predict the output without running the code. Then check your answer in Python.

```
x = 8
y = 3
print(x + y)
print(x * y)
print(x / y)
print(x % y)
```

2. Write a Python expression that converts a length provided by the user in inches to meters.

2 Strings and Input

1. Create a program that asks the user for a name and an age, and prints the results in the following format using an f-string: "name: age."

3 Lists and Tuples

1. Consider

```
student = ("Maya", 20681001, 4)
course = ["Calculus",["Maya", "Liam", "Ava"]]
```

What is produced by each expression?

```
student[1]
student[2]
course[0]
course[1][2]
```

2. Add `"Noah"` to the student list inside `course`.

4 Boolean Logic and Decisions

1. Let

```
score = 3
submitted = True
```

Predict the value of each expression:

```
score >= 4
score >= 2 and submitted
score == 0 or score == 1
not submitted
```

2. Complete the decision:

```
if score >= 4:
    print("Calculus")
elif __________:
    print("Trigonometry")
else:
    print("Algebra")
```

5 Loops

1. What values are printed?

```
for i in range(4):
    print(i)
```

How many times does the loop execute?

2. Complete the following loop so that it prints integers in order:

```
i = 0
while i < 4:
    print(i)
    __________
```

3. Consider

```
scores = [5, 1, 3, 4, 2]
```

Write a `for` loop that prints `"Calculus"` whenever the score is 4 or 5.

6 Challenge

1. Define the following *diagonal* matrix:

```
A = [
    [1, 0, 0],
    [0, 1, 0],
    [0, 0, 1]
]
```

Suppose we did not know the matrix was diagonal. Create nested `for` loops and conditional logic that demonstrate the matrix is diagonal. For example:

```
is_diagonal = True
for row in range(2):
    for column in range(3):
        if(...):
         is_diagonal = False
         ...
```